The U.S. Debt Valuation Puzzle (2024)

Can U.S. Government Debt Be Justified by Future Fiscal Surpluses?

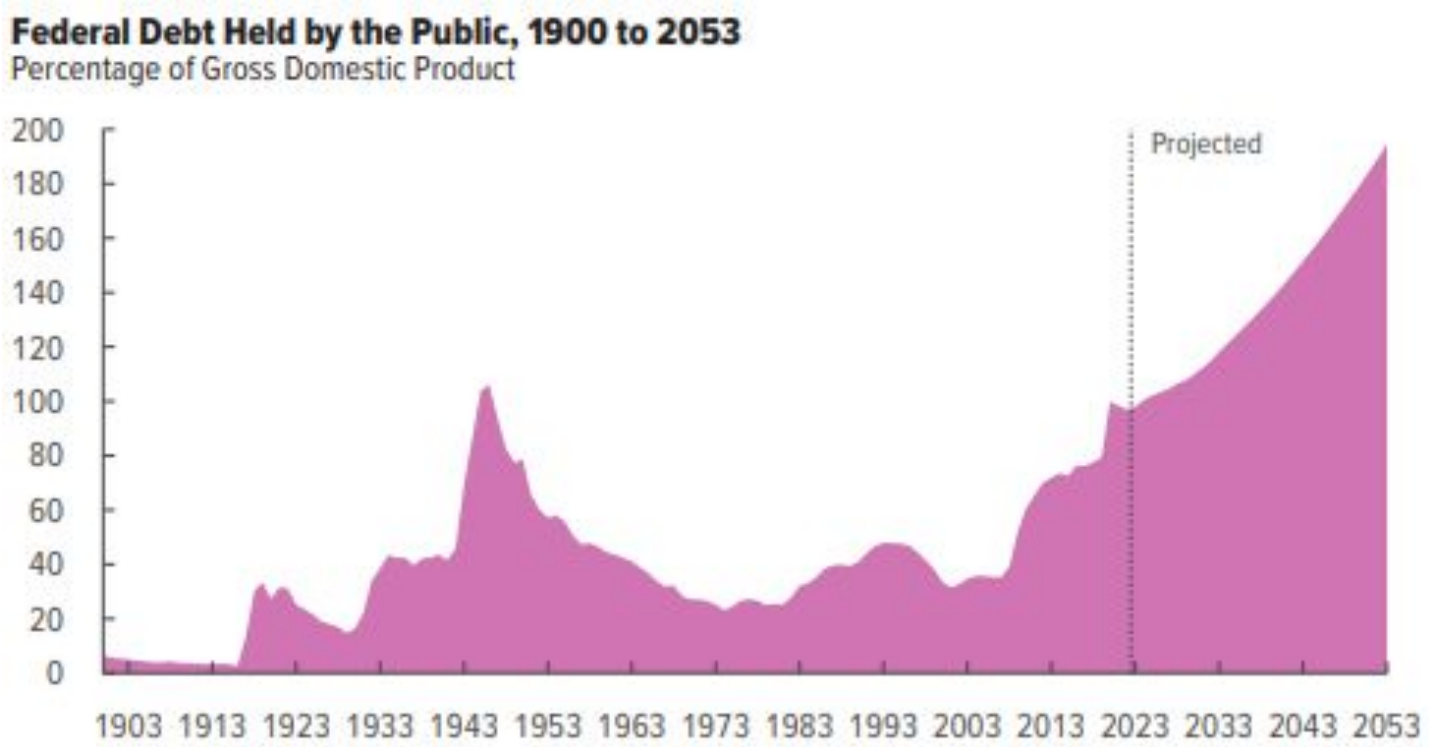
By : Jiang, Lustig, Nieuwerburgh , Xiaolan

Presented by : Damien Levacher & Japhet Ayassou

Flow of the Presentation

1. Overview of the paper
2. A new approach
3. Future surpluses pricing
4. What is the valuation puzzle?
5. Cointegrating check
6. Two explanations for the valuation gap
7. Fixes to fix the valuation gap : Seigniorage, Risk Premia, Peso Problem
8. Conclusion: The Valuation Puzzle Remains Open

U.S. Debt Is on an Unprecedented Trajectory



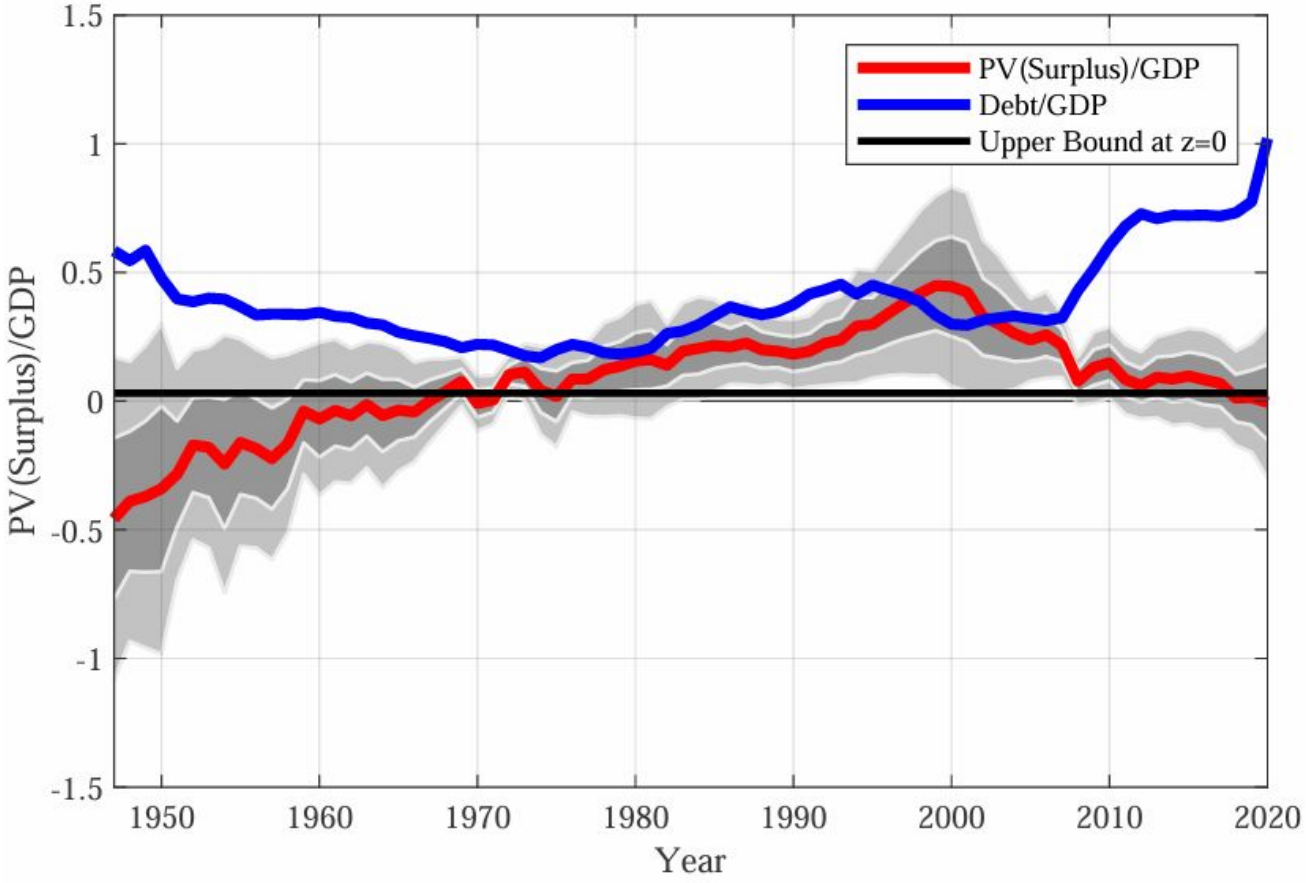
Overview

- Blanchard (2019): When r < g, the government may sustain deficits without fiscal crises — debt need not be backed dollar-for-dollar by surpluses
- This paper challenges that claim
- It asks: **Is the debt backed by the future surpluses ?**
- Their answer is : NO!
- The PDV of future surpluses is actually negative

The approach : 03 novelties

- All of the outstanding US bonds considered together :
  - Studies before focus only on one single bond
  - This paper is the first to consider all of them together as one asset
- Surplus risk integrated :
  - Paper before only use interest rate to compute the PV of future surplus
  - This is the first one a factor (SDF) that takes into account that risk (in addition to the interest)
- Same SDF for pricing both bonds and stocks : no-arbitrage

What Is the Valuation Puzzle?



PDV future surpluses computation

➢ The pricing equation :

PDV surpluses = $E_t \left[\sum_{j=0}^{\infty} M_{t,t+j} \left(T_{t+j} - G_{t+j} \right) \right]$

E[XY] = E[X] x E[Y] + Cov(X, Y)

$$= \sum_{j=0}^{T} P_t^{\$}(j) \mathbb{E}_t \left[S_{t+j} \right] + \sum_{j=0}^{T} \mathrm{Cov}_t \left(M_{t,t+j}^{\$}, T_{t+j} \right) - \sum_{j=0}^{T} \mathrm{Cov}_t \left(M_{t,t+j}^{\$}, G_{t+j} \right)$$

TVC is assumed. And by no-arbitrage : Price today (j) = E[M_t,t+j]

Finite version $\sum_{j=0}^{T} P_t^{\$}(j) \mathbb{E}_t \left[S_{t+j} \right] + \sum_{j=0}^{T} \mathrm{Cov}_t \left(M_{t,t+j}^{\$}, T_{t+j} \right) - \sum_{j=0}^{T} \mathrm{Cov}_t \left(M_{t,t+j}^{\$}, G_{t+j} \right) + \mathbb{E}_t \left[M_{t,t+T} D_{t+T} \right]$

What Is the Valuation Puzzle?

- Theoretically, what we should observe : D_t = PDV_t surplus

- But in reality : $D_t > E_t \left[\sum_{j=0}^{\infty} M_{t,t+j} \left(T_{t+j} - G_{t+j} \right) \right]$

- The goal of the paper is to find what's going on exactly ?
- What are the reasons investors continue to lend to the US ?

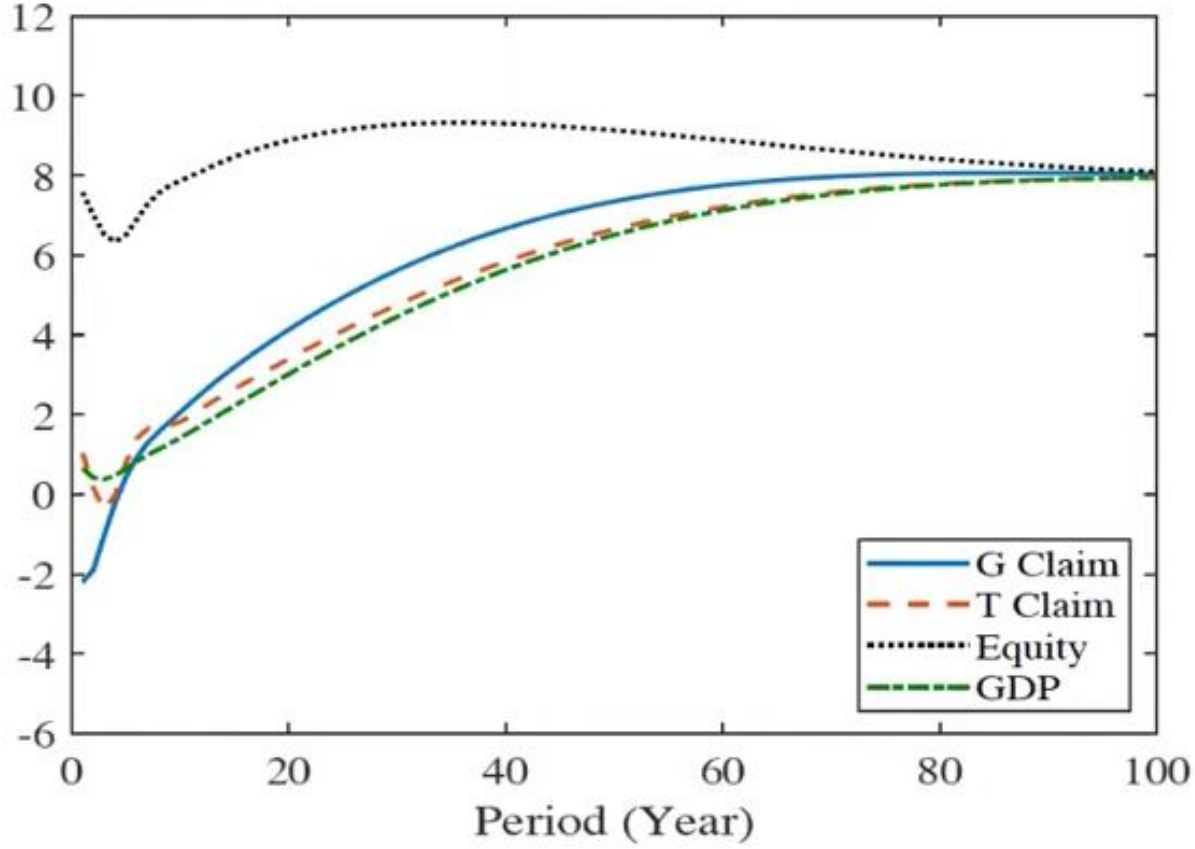
Empirical Validation: Taxes, Spending, and GDP Are Cointegrated

- We estimate a Vector Error Correction Model (VECM) in log(T), log(G), log(GDP)

$$\Delta w_t = A\left(B' w_{t-1} + c\right) + D\Delta w_{t-1} + \varepsilon_t, \quad \text{where } w_t = \begin{pmatrix} \log T_t \\ \log G_t \\ \log \mathrm{GDP}_t \end{pmatrix}.$$

- Johansen test: Reject rank 0 ($p = 0.01$); cannot reject rank 1 ($p = 0.11$) or rank 2 ($p = 0.96$) → cointegration confirmed
- Taxes and GDP share a permanent stochastic trend, leading to a permanent shock to GDP permanently shifts future taxes. Their long-run risk premia must be equal — and both exceed the long-term real bond yield.

Long-Run Convergence



Main Result: Two Explanations for the Valuation Gap

➢ **Finding:** The PV of future surpluses consistently falls below the market value of U.S. debt — this is the valuation puzzle

- **Explanation 1 — Investor Beliefs:** Investors may systematically overestimate future surpluses (expectational optimism), inflating the debt's perceived backing
- **Explanation 2 — Convenience Yields:** U.S. Treasuries provide a *liquidity/safety premium* that investors are willing to pay above fundamental value — this yield also backs the debt (Proposition 4)

Even accounting for both channels, the puzzle is not fully resolved

Trying to fix the Valuation Gap

- **Convenience Yield / Seigniorage** : To close the gap the when the dt ≈0.34 then λt≈6%, ten times bigger than our average convenience yield λ0 = 0.56%

$$(\tau_0 + d_0\lambda_0)\exp(pd_0^{\tau}) - g_0\exp(pd_0^{g})$$

- **Risk Premium** : The gap persists whether GDP risk premium is 3% or 2.5%

$$\left(\frac{PV_t^s}{Y_t}\right)^u(z=0) = \exp(pd_0^y)(\tau_0 - g_0)\; m_{t+1}^{\$} = -y_t^{\$}(1) - \frac{1}{2}\Lambda_t'\Lambda_t - \Lambda_t'\varepsilon_{t+1},\quad \Lambda_t = \Lambda_0 + \Lambda_1 z_t$$

- **Peso Problem** : rare fiscal catastrophe where to close the gap should happen every 3 years where in the equation the parameters needs to be at ϕ(probability for the event to happen)= 36%, ℓ = 30%(size of spending cut

$$P_t^{\text{Cut,Peso}} \stackrel{\text{def}}{=} P_t^G - P_t^{G,\text{Peso}} = \mathbb{E}_t\left[M_{t+1}^{\$}\left(\phi_t\left(\ell\cdot\left(G_{t+1}+P_{t+1}^G\right)\right) + (1-\phi_t)P_{t+1}^{\text{Cut,Peso}}\right)\right].$$

All of them widened the gap even more or became impossible events

Conclusion: The Valuation Puzzle Remains Open

Blanchard (2019): $r < g \Rightarrow$ debt may be sustainable

↓

Data: Debt/GDP $\gg$ PV(Surplus)/GDP consistently

↓

Channels tested: Convenience yields + Premium Risk + Peso Problem

↓

Result: Gap remains unexplained $\Rightarrow$ The Valuation Puzzle

↓

Possible explanations: Convenience yields are underestimated, or investors are optimistic

↓

Open questions: Bubble? Reserve currency premium? Structural factors?